Basic Statistics

Exercise

If X is binomially distributed with 6 trials and a probability of success equal to $\frac{1}{4}$ at each attempt, what is the probability of:

(a) exactly 4 successes (b) at least one success?

Exercise

Solution:

Exercise2

When an unbiased coin is tossed eight times what is the probability of obtaining:

(a) less than 4 heads (b) more than five heads?

Exercise2

Solution:

Exercise3

A biased die is thrown thirty times and the number of sixes seen is eight. If the die is thrown a further twelve times find:

(a) the probability that a six will occur exactly twice;

(b) the expected number of sixes;

(c) the variance of the number of sixes.

Exercise3

Solution:

Exercise4

A random variable X is binomially distributed with mean 6 and variance 4.2. Find $P(X \leq 6)$.

Exercise4

Solution:

Exercise5

A pewter manufacturer produces souvenir mugs. Suppose that one of the machines breaks down and 8% of the mugs are found to be defective and cannot be sold. If 23 mugs are selected at random, find the probability that

a) 3 mugs are defective

b) Between 8 to 10 mugs are defective

c) At least 1 mug cannot be sold.

Exercise5

Solution:

Exercise6

1. Suppose Z is a standard normal random variable. What is $P(Z \leq 1.2)$?

2. Suppose Z is a standard normal random variable. What is $P(Z \leq -2.36)$?

3. Suppose Z is a standard normal random variable. What is $P(Z \leq -0.41)$

4. Suppose Z is a standard normal random variable. What is $P(-0.41 \leq Z \leq 1.2)$?

5. Suppose Z is a standard normal random variable. What is $P(Z > 1.96)$?

Exercise6

Solution:

Exercise7

The dressed weights of Excelsior Chickens are approximately normally distributed with mean 3.20 pounds and standard deviation 0.40 pound. About what proportion of the chickens have dressed weights greater than 3.60 pounds?

Exercise7

Solution:

Exercise8

A cooper test for a football player from Team A is normally distributed with mean 660sec and standard deviation 45sec. The Cooper test for a football player from Team B is normally distributed with mean 690sec and std 25sec. A player is selected at random.

a) What is the probability a player from Team A can complete the test less than 700sec?

b) What is the probability the time set by a Team A player is better than the time set by a Team B player?